Control Theory

Chap 2: Model and Dynamics

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Objectives:

1. Know the basic methods on modelling electrical, mechanical system and even space vehicle;
2. Can transform the frequency and time domain models, and calculate the system state under the homogeneous and non-homogeneous situations;
3. Can apply Mason's law to build the Laplace transfer function with the block diagram, signal flow graphs;
4. Comprehend the importance and meaning of Roots;
5. Know the synthesis process of a control plant to its dynamic models in frequency and time domains;
6. Able to analyze and compare the transient and steady-state performance of closed and open loop systems.



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Chap 2: Model and Dynamics

1. Model in Differential Equation
2. Model in Laplace Equation and Transfer Function
3. Block Diagram - Signal-flow Graph- Mason's Rule
4. Model in State-space Equation
5. Nonlinear Model and Linearization
6. Transient and Steady-State Responses

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1 Model in Differential Equation

2 Model in Laplace Equation and Transfer Function

3 Block Diagram - Signal-flow Graph- Mason's Rule

4 Model in State-space Equation

5 Nonlinear Model and Linearization

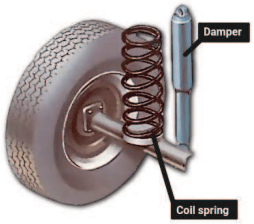
6 Transient and Steady-State Responses

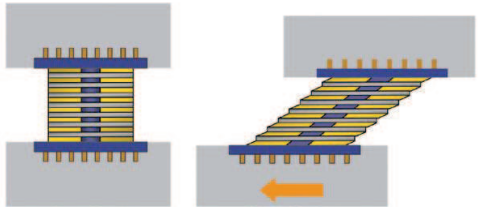
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1 Model in Differential Equation
- Mechanical Spring-damper-mass System
- Inductor Resistor Capacitor -LRC and Electrical Systems
- Satellite Attitude and Orbit
- Cart Inverted Pendulum System
- Electrical Motor

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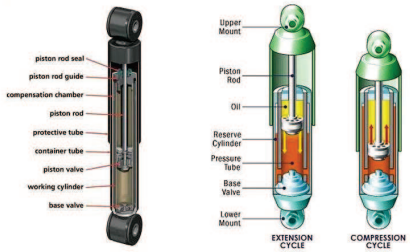
Consider the spring-damper-mass system



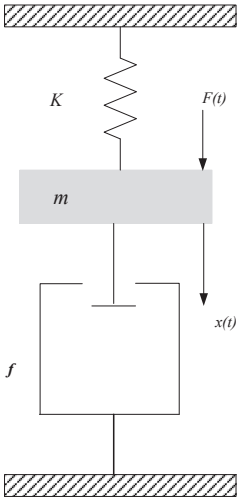


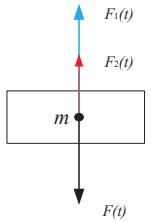


Appollo 11 Suspension Component



Vehicle Suspension Component





Via Newton's law, Spring and Damper forces

$$\begin{cases} m\ddot{x}(t) + mg &= F(t) - F_1(t) - F_2(t) \\ F_1(t) &= Kx(t) \\ F_2(t) &= D\dot{x}(t) \end{cases}$$

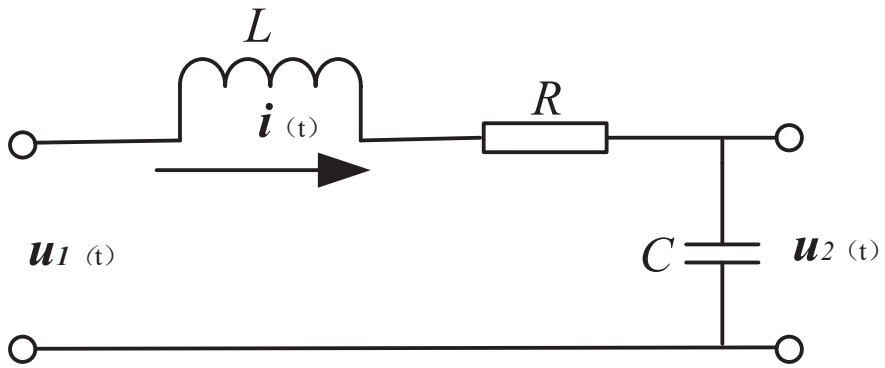
one has that

$$F(t) = m\ddot{x}(t) + mg + Kx(t) + D\dot{x}(t)$$

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Consider the single-input-single-output (SISO) electrical circuit



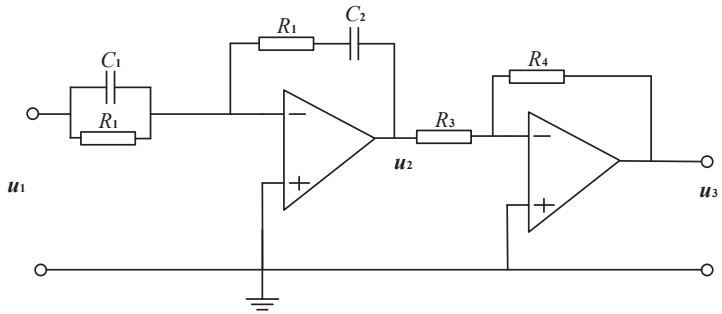
Based on Kirchhoff's current and voltage laws, one has

$$\begin{aligned} u_1(t) - u_2(t) &= L\frac{di(t)}{dt} + Ri(t) \\ u_2(t) &= \frac{1}{C}\int i(\tau)d\tau \end{aligned}$$

Eliminating the variable of current $i(t)$, it verifies

$$u_1(t) = LC\ddot{u}_2(t) + RC\dot{u}_2(t) + u_2(t)$$

Electrical system with operational amplifiers



For input u_1 and output u_2 of the left isolation operator amplifier

$$\begin{aligned} \frac{1}{C_1}\int i_{c1}(\tau)d\tau &= u_{2i}-u_1 \\ i_{R1}R_1 &= u_{2i}-u_1 \\ \frac{1}{C_2}\int i_{c2}(\tau)d\tau &= u_2-u_{2i} \\ i_{R2}R_2 &= u_2-u_{2i} \end{aligned}$$

the input current equals the output current

$$i_{c1}(t)+i_{R1}(t)=i_{c2}(t)+i_{R2}(t)$$

it can be further verified that

$$C_1\left(\dot{u}_{2i}-\dot{u}_1\right)+\frac{u_{2i}-u_1}{R_1}=C_1\left(\dot{u}_2-\dot{u}_{2i}\right)+\frac{u_2-u_{2i}}{R_2}$$

with $u_{2i}=0$, one can further verify that

$$-C_1\dot{u}_1-\frac{u_1}{R_1}=C_1\dot{u}_2+\frac{u_2}{R_2} \tag{1}$$

for the input u_2 and output u_3 of the right isolation operator amplifier,

$$\frac{u_3}{R_4}=\frac{u_2}{R_3} \tag{2}$$

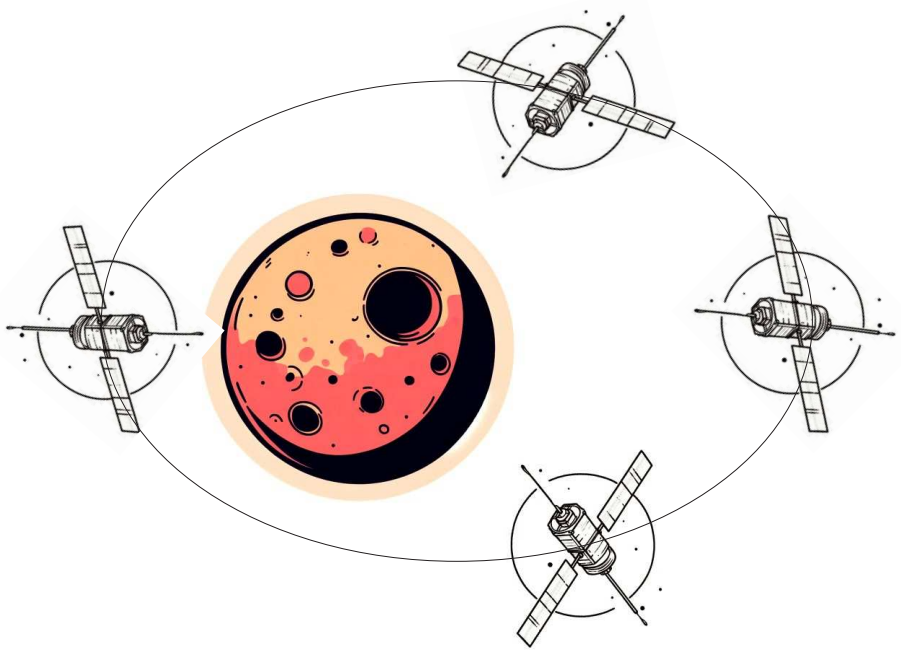
Combining (1) and (2), it verifies that

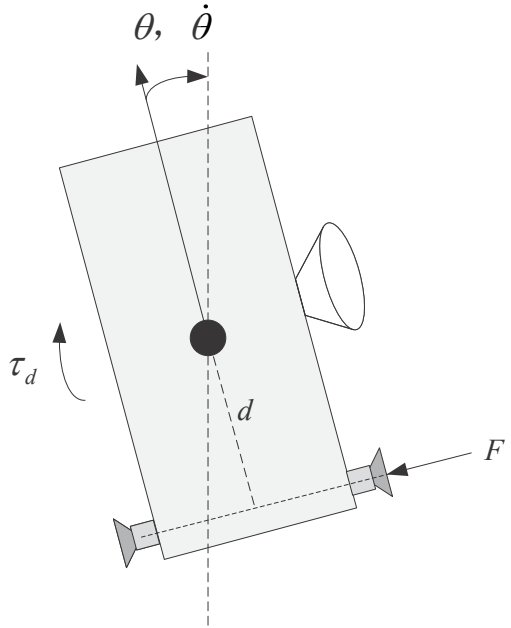
$$-C_1\dot{u}_1-\frac{u_1}{R_1}=C_1\frac{R_3}{R_4}\dot{u}_3+u_3\frac{R_3}{R_2R_4} \tag{3}$$

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A simple satellite flying around Mars

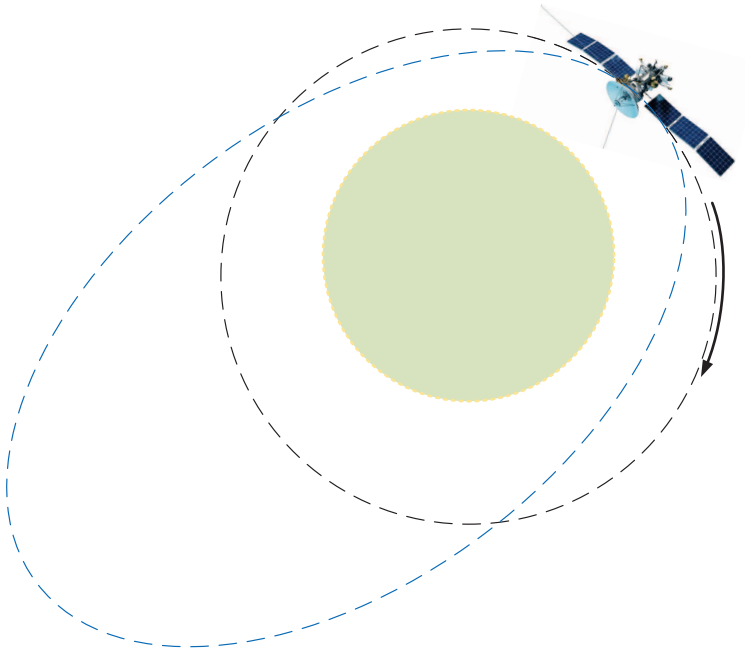




The rotation / attitude model of the satellite is

$$I\ddot{\theta} = Fd + \tau_d \tag{4}$$

Consider typical two-body system with unbiased circular orbit, like satellite around the sun, earth or other celestial body.
circular or elliptical orbits



According to Newton's second law

$$m\ddot{\mathbf{r}} = -\frac{GMm}{r^2}\frac{\mathbf{r}}{r}$$

Further considering the orbit perturbation and external disturbance, one has the dynamics of satellite in the inertial coordinates,

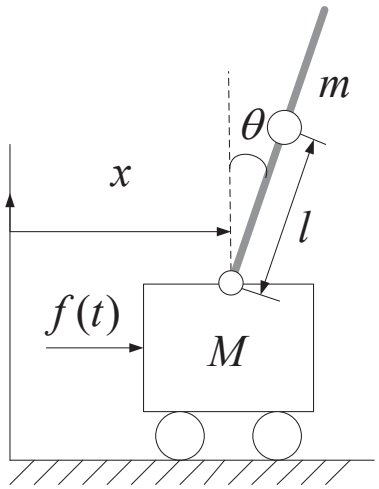
$$\begin{cases} \ddot{x} &= -\mu\dfrac{x}{r^3} + g_{px} + \dfrac{F_x}{m} \\ \ddot{y} &= -\mu\dfrac{y}{r^3} + g_{py} + \dfrac{F_y}{m} \\ \ddot{z} &= -\mu\dfrac{z}{r^3} + g_{pz} + \dfrac{F_z}{m} \\ \dot{m} &= -\dfrac{F}{I_{sp}g_0} \end{cases}$$

m	the satellite mass including the fuel (liquid or solid)	x, y and z	position of space satellite in the inertial coordinates
g_0	the standard acceleration due to gravity.	$\mathbf{g_p} = [g_{px}\ g_{py}\ g_{pz}]$	perturbing accelerations
μ	standard gravitational parameter due to the celestial body $\mu \equiv GM$	F_x, F_y and F_z	thrust components in the inertial frame
I_{sp}	the specific impulse of the spacecraft engine	F	the thrust
M	the mass of the celestial body	G	the gravitational constant

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Driven-less inverted-pendulum, motor-driven cart. Force $f(t)$ applies to the cart. θ: pendulum to the vertical line angle.



The rotational motion of the pendulum rod

$$I\ddot{\theta} = Vl\sin(\theta) - Hl\cos(\theta) \tag{5}$$

I is the moment of inertia of the rod about its center of gravity. The horizontal dynamics is

$$m\frac{d^2}{dt^2}\left(x + l\sin(\theta)\right) = H \tag{6}$$

The vertical motion of pendulum rod is

$$m\frac{d^2}{dt^2}\left(l\cos(\theta)\right) = V - mg \tag{7}$$

where H and V are artificially introduced intermediate variables. The cart horizontal motion is

$$f - H = M\ddot{x} \tag{8}$$

Therefore, with (6) and (8), one has that

$$f - m\frac{d^2}{dt^2}\left(x + l\sin(\theta)\right) = M\ddot{x} \tag{9}$$

and, with (5), (7), and (8), one has that

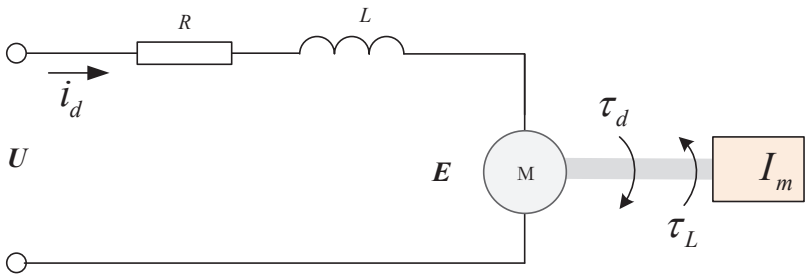
$$I\ddot{\theta} = \left(m\frac{d^2\left(l\cos(\theta)\right)}{dt^2} + mg\right) l\sin(\theta) - \left(f - M\ddot{x}\right) l\cos(\theta) \tag{10}$$

Intermediate variables V and H are eliminated.

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U : driven source voltage, i_d : driven current, R and L are the electrical motor resistor and inductor. τ_d and τ_L : the driven and load torques. I_m: load inertia. k_f friction coefficient of load mass.



For the input voltage, one has that

$$U = i_d R + L\frac{di_d}{dt} + E \tag{11}$$

τ_d is proportional to current i_d, whose coefficient is k_d, which means

$$\tau_d = k_d i_d \tag{12}$$

reaction voltage E is proportional to the rotation velocity ω_n, the coefficient is k_n, it verifies

$$E = k_n \omega_n \tag{13}$$

The inertia of the rotation is I_m,

$$I_m \dot{\omega}_n + k_f \omega_n = \tau_d - \tau_L \tag{14}$$

Via (11) and (13), one has

$$U = i_d R + L\frac{di_d}{dt} + k_n \omega_n \tag{15}$$

By (12) and (14), it verifies

$$I_m \dot{\omega}_n + k_f \omega_n = k_d i_d - \tau_L \tag{16}$$

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Laplace Transformation

$$F(s) = \mathfrak{L}(f(t)) = \int_0^{+\infty} f(\tau)e^{-s\tau}d\tau \qquad (17)$$

Inverse Laplace Transformation

$$f(t) = \mathfrak{L}^{-1}(F(s)) = \frac{1}{2\pi j}\int_{c-j\infty}^{c+j\infty} F(s)e^{st}ds \qquad (18)$$

Theorem (Linear Nature Theorem)

For two Laplace functions $F_1(s)$ and $F_2(s)$, if

$$\mathfrak{L}^{-1}F_i(s) = f_i(t), \text{ for } i = 1, 2. \qquad (19)$$

then

$$\mathfrak{L}(af_1(t) + bf_2(t)) = aF_1(s) + bF_2(s) \qquad (20)$$

Theorem (Frequency-shift Theorem)

For the time domain function $f(t)$, whose Laplace transformation is $F(s)$, then for another time domain function $e^{-at}f(t)$, one has that

$$\mathfrak{L}\left[e^{-at}f(t)\right] = F(s+a) \qquad (21)$$

Theorem (Differentiation of Laplace Transform)

If $X(s)$ is the Laplace transform of $x(t)$, then

$$\frac{dX(s)}{ds} = \mathfrak{L}(-tx(t)) \qquad (22)$$

More generally,

$$\frac{d^nX(s)}{ds^n} = \mathfrak{L}\left((-t)^n x(t)\right) \qquad (23)$$

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Theorem (Transform of Derivatives)

For a function $x(t)$, whose Laplace Transformation is $X(s)$, then the derivative of function $x(t)$ is

$$\mathfrak{L}\left(\frac{dx(t)}{dt}\right) = sX(s) - X(0) \tag{24}$$

More generally, if all the derivatives and limits exist and are right-continuous,

$$\mathfrak{L}\left(\frac{d^n x(t)}{dt^n}\right) = s^n X(s) - x(0)s^{n-1} - \dot{x}(0)s^{n-2} - \frac{d^{n-2}x}{dt^{n-2}}(0)s - \frac{d^{n-1}x}{dt^{n-1}}(0) \tag{25}$$

Theorem (Positive Time Shift)

Suppose that

$$X(s) = \mathfrak{L}(h(t)x(t)) \tag{26}$$

For any positive time τ , the transform of the function x shifted by time τ is

$$\mathfrak{L}\left(h(t-\tau)x(t-\tau)\right) = e^{-\tau s}X(s) \tag{27}$$



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Laplace Transformation Examples

Example

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

$$\mathcal{L}\{u(t)\} = \int_0^\infty e^{-st} \cdot 1 \, dt = \left[-\frac{e^{-st}}{s} \right]_0^\infty = \frac{1}{s}$$

Example

$$\begin{aligned} \mathcal{L}\{e^{at}\} &= \int_0^\infty e^{-st} e^{at} \, dt \\ &= \int_0^\infty e^{-(s-a)t} \, dt \\ &= \left[-\frac{e^{-(s-a)t}}{s-a} \right]_0^\infty \\ &= \frac{1}{s-a} \end{aligned}$$

Example

$$\begin{aligned} \mathcal{L}\{\sin(\omega t)\} &= \frac{1}{2i} \left(\mathcal{L}\{e^{i\omega t}\} - \mathcal{L}\{e^{-i\omega t}\} \right) \\ &= \frac{1}{2i} \left(\frac{1}{s - i\omega} - \frac{1}{s + i\omega} \right) \\ &= \frac{1}{2i} \cdot \frac{2i\omega}{s^2 + \omega^2} = \frac{\omega}{s^2 + \omega^2} \end{aligned}$$

Example

$$\frac{c}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b}$$

where

$$A + B = 0, Ab + Ba = c$$

Example

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} e^{st}\frac{1}{s}\,ds, \gamma > 0$$

$$\text{Res}\left(\frac{e^{st}}{s}, 0\right) = \lim_{s\to 0} s\cdot\frac{e^{st}}{s} = e^0 = 1$$

$$\frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty}\frac{e^{st}}{s}\,ds = 1$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = \begin{cases}1 & t>0\\ 0 & t<0\end{cases} = u(t)}$$

Example

$$\mathcal{L}^{-1}\left\{\frac{\omega}{s^2+\omega^2}\right\} = \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} e^{st}\frac{\omega}{s^2+\omega^2}\,ds$$

$\gamma > 0$

Example

$$\begin{cases}\text{Res}\left(\frac{\omega e^{st}}{s^2+\omega^2}, i\omega\right) = \lim_{s\to i\omega}(s-i\omega)\cdot\frac{\omega e^{st}}{s^2+\omega^2}\\ = \frac{\omega e^{i\omega t}}{2i\omega} = \frac{e^{i\omega t}}{2i}\\ \text{Res}\left(\frac{\omega e^{st}}{s^2+\omega^2}, -i\omega\right) = \lim_{s\to -i\omega}(s+i\omega)\cdot\frac{\omega e^{st}}{s^2+\omega^2}\\ = \frac{\omega e^{-i\omega t}}{-2i\omega} = -\frac{e^{-i\omega t}}{2i}\end{cases}$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{\omega}{s^2+\omega^2}\right\} = \sin(\omega t)}$$

Example

$$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = \frac{1}{2\pi i}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} e^{st}\frac{1}{s-a}\,ds$$

$$\boxed{\frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty}\frac{e^{st}}{s-a}\,ds = e^{at}}$$

2 Model in Laplace Equation and Transfer Function

- Laplace and Inverse Laplace Transformation Definition
- Rational Laplace Transforms
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Laplace transform $F(s)$ of a function $f(t)$ is a rational function of s if it can be written as a ratio of polynomials in s as

$$F(s) = \frac{N(s)}{D(s)} \tag{28}$$

where $N(s)$ and $D(s)$ are polynomials of the complex variable s as

$$\textbf{Numerator: } N(s) = \sum_{i=0}^{m} b_i s^i = b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s^1 + b_0 \tag{29}$$

$$\textbf{Denominator: } D(s) = s^n + \sum_{i=0}^{n-1} a_i s^i = s^n + a_{n-1} s^{n-1} + \cdots + a_1 s^1 + a_0 \tag{30}$$

In (29) and (30), the coefficients $b_m, b_{m-1}, \cdots, b_1, b_0$ and $a_{n-1}, \cdots, a_1, a_0$. Parameters m and n are positive integers.
The degree of $D(s)$, n , is the order of the rational function $F(s)$.

Definition (Proper Rational, Strict rational)

$F(s)$ is a *proper rational function* when $m \leq n$, and, *strict rational function* when $m < n$.

Definition (Zeros and Poles)

$z_1, z_2, \cdots, z_m$ are the *zeros* of $F(s)$ when $s = z_i$ for $i = 1, 2, \cdots, m$ by setting $F(s) = 0$;
$p_1, p_2, \cdots, p_n$ are the *poles* of $F(s)$ when s approaches p_i for $i = 1, 2, \cdots, n$, the magnitude $|F(s)|$ becomes ∞.
the rational function

$$F(s) = \frac{N(s)}{D(s)} = K\frac{(s-z_1)(s-z_2)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)} \tag{31}$$

Case 1: distinct poles
If $F(s)$ is strictly proper $(n > m)$ and the poles $p_1, p_2, \cdots, p_n$ of $F(s)$ are distinct (nonrepeated), then $F(s)$ has the partial fraction expansion below

$$F(s) = \sum_{i=1}^{n} \frac{c_i}{s-p_i} = \frac{c_1}{s-p_1} + \frac{c_2}{s-p_2} \cdots + \frac{c_{n-1}}{s-p_{n-1}} + \frac{c_n}{s-p_n} \tag{32}$$

where the parameters c_i are the residues can be given as

$$c_i = [(s-p_i)F(s)]_{s=p_i}, \text{for } i = 1, 2, \cdots, n \tag{33}$$

- The residue c_i is real if and only if the corresponding pole p_i is real
- c_i is complex if and only if p_i is complex.

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For (32), via inverse Laplace transform definition, one has the inverse transformation of $F(s)$ that

$$f(t) = \mathfrak{L}^{-1}(F(s)) = \sum_{i=1}^{n} c_i e^{p_i t} = c_1 e^{p_1 t} + c_2 e^{p_2 t} + \cdots + c_{n-1} e^{p_{n-1} t} + c_n e^{p_n t} \tag{34}$$

If $F(s)$ has a pair of complex poles $p = \sigma \pm j\omega$, then $f(t)$ contains the term

$$ce^{\sigma t + j\omega t} + c^* e^{\sigma t - j\omega t} \tag{35}$$

$c^* = c_r - jc_j$ is the complex conjugate of $c = c_r + jc_j$. Complex coefficient parameter c can be

$$c = c_r + jc_j = |c|e^{j\theta} \tag{36}$$

$|c| = \sqrt{c_r^2 + c_j^2}$, $\tan(\theta) = \frac{c_j}{c_r}$. By Euler' s formula $e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$, one can obtain that

$$\begin{aligned} ce^{\sigma t + j\omega t} + c^* e^{\sigma t - j\omega t} &= |c|e^{j\theta} e^{\sigma t + j\omega t} + |c^*|e^{-j\theta} e^{\sigma t - j\omega t} \\ &= |c|e^{\sigma t}(e^{j(\theta+\omega t)} + e^{-j(\theta+\omega t)}) \\ &= 2|c|e^{\sigma t}\cos(\omega t + \theta) \end{aligned} \tag{37}$$

If $F(s)$ has a pair of complex poles, then $f(t)$ contains a sinusoidal term with an exponential amplitude factor $e^{\sigma t}$. When $\sigma = 0$, which means the poles are purely imaginary, (37) is pure sinusoidal.

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Case 2: Real pole at $s = p$ repeated r times

If one of the poles (for example p_1) is repeated r times and the other $n - r$ poles are distinct, $F(s)$ has the partial fraction expansion

$$F(s) = \frac{c_1}{s - p_1} + \frac{c_2}{(s - p_1)^2} + \cdots + \frac{c_r}{(s - p_1)^r} + \frac{c_{r+1}}{s - p_{r+1}} + \cdots + \frac{c_n}{s - p_n} \tag{38}$$

and the residues $c_1, c_2, \cdots, c_r$ can be calculated as

$$c_i = \frac{1}{(r - i)!} \left\{ \frac{d^{r-i}}{ds^{r-i}} \left[(s - p_1)^r F(s) \right] \right\}_{s=p_1} \tag{39}$$

The inverse Laplace transform would be

$$f(t) = \mathfrak{L}^{-1} \left(F(s) \right) = c_1 e^{p_1 t} + c_2 t e^{p_1 t} + \cdots + \frac{c_r}{(r-1)!} t^{r-1} e^{p_1 t} + c_{r+1} e^{p_{r+1} t} \cdots + c_n e^{p_n t} \tag{40}$$



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When $m = n$ in (28), which means $F(s)$ is not strictly proper, it has that

$$F(s) = \frac{N(s)}{D(s)} = b_n + \frac{R(s)}{D(s)} \tag{41}$$

Degree of $R(s)$ is strictly less than n. Inverse transformation of (41), $f(t)$, contains impulse function $b_n\delta(t)$.

Table 1: Relationship between the Poles of $F(s)$ and the Form of $f(t)$

Pole Locations of $F(s)$	Corresponding Terms in $f(t)$
Nonrepeated real pole at $s = p$	ce^{pt}
Real pole at $s = p$ repeated r times	$\sum_{i=1}^{r} c_i t^{i-1} e^{pt}$
Nonrepeated complex pair at $s = \sigma \pm j\omega$	$ce^{\sigma t}\cos(\omega t + \theta)$
Complex pair at $s = \sigma \pm j\omega$ repeated r times	$\sum_{i=1}^{r} c_i t^{i-1} e^{\sigma t}\cos(\omega t + \theta)$



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Open Left Half-plane

The poles of Laplace function $F(s)$ determine the convergence of the time-domain function $f(t)$, when the real parts of all the poles $p_1, p_2, \cdots, p_n$ of $F(s)$ are strictly negative (i.e. $Re(p_i) < 0$ for $i = 1, 2, \cdots, n.$), $f(t)$ would converge to zero when $t \to \infty$. It means all the poles locate in the open left half-plane (OLHP), which is the region in the complex plane to the left of the imaginary axis.

Theorem (Final Value Theorem)

When all the poles of $F(s)$ locate in the left region of the imaginary axis, except that $F(s)$ may have any nonrepeated pole at $s = 0$, the function $f(t)$ would have a finite limit $f(t)$ as $t \to \infty$. It can be expressed as

$$f(\infty) = \lim_{t\to\infty} f(t) = \lim_{s\to 0} [sF(s)] \text{ , if } Re(p_i) < 0 \text{ for all poles } p_i \neq 0 \tag{42}$$

Steady state error calculation method！ ！ ！



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Example

Laplace decomposition

$$\begin{aligned} f(s) &= \frac{a-b}{(s-a)(s-b)}, a \neq b \neq 0 \\ f(s) &= \frac{cs+d}{s(s-a)^2}, a \neq 0 \end{aligned} \tag{43}$$

Example

Calculate the final value when $t \to \infty$

$$\begin{aligned} f(t) &= t^2 - 2t + 1 \\ f(t) &= te^{at}, a > 0 \\ f(t) &= te^{-at}, a > 0 \\ f(t) &= be^{-at}(\sin(\omega t + \frac{\pi}{4})), a > 0, b \in \Re, \text{ and } \omega \in \Re. \end{aligned} \tag{44}$$



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2 Model in Laplace Equation and Transfer Function

- Laplace and Inverse Laplace Transformation Definition
- Rational Laplace Transforms
- Linear Differential Equation Solution

Example

Consider the second-order differential equation, and the initial conditions

$$\ddot{y} + a\dot{y} + by = u(t), y(0) = y_0, \quad \dot{y}(0) = y_{d0} \tag{45}$$

Taking the Laplace transform of both sides and using the transform properties, one has

$$\left(s^2 Y(s) - sy_0 - y_{d0}\right) + a\left(sY(s) - y_0\right) + bY(s) = U(s) \tag{46}$$

rewrite (46) as

$$(s^2 + as + b)Y(s) - sy_0 - y_{d0} - ay_0 = U(s) \tag{47}$$

and $Y(s)$ can be verified as

$$Y(s) = \frac{U(s)}{s^2 + as + b} + \frac{sy_0 + y_{d0} + ay_0}{s^2 + as + b} \tag{48}$$

Via the inverse Laplace transformation, one can have the time-domain solution $y(t)$

$$y(t) = \mathfrak{L}^{-1}\left(Y(s)\right) = \underbrace{\mathfrak{L}^{-1}\left(\frac{U(s)}{s^2 + as + b}\right)}_{\text{nonhomogeneous part: Input Response}} + \underbrace{\mathfrak{L}^{-1}\left(\frac{sy_0 + y_{d0} + ay_0}{s^2 + as + b}\right)}_{\text{homogeneous part: Initial State Response}} \tag{49}$$

Example

Consider the mechanical damper spring system whose differential model is

$$f(t) = m\ddot{x}(t) + Kx(t) + D\dot{x}(t) \tag{50}$$

if the initial state $x(0) = 0$, and its differential $\dot{x}(0) = 0$, by the theorems given in this section one can obtain its Laplace transform model below

$$F(s) = s^2 mX(s) + KX(s) + sDX(s) \tag{51}$$

then one can obtain the transfer function from input $F(s)$ to output $X(s)$ below

$$\frac{X(s)}{F(s)} = \frac{1}{s^2 m + sD + K} \tag{52}$$

where $F(s)$ and $X(s)$ are the Laplace transformation of $f(t)$ and $x(t)$.



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Example

Consider the electrical motor whose differential model equations can be expressed

$$\begin{cases} U = i_d R + L\dfrac{di_d}{dt} + k_n \omega_n \\ I_m \dot{\omega}_n + k_f \omega_n = k_d i_d - \tau_L \end{cases} \tag{53}$$

if the outputs and their derivatives are zero, then for the two inputs $u(t)$ and $\tau_L(t)$ and the two outputs $\omega_n(t)$ and $i_d(t)$, one has the following Laplace-transformed model

$$\begin{cases} U(s) = I_d(s)R + sLI_d(s) + k_n \Omega_n(s) \\ sI_m \Omega_n(s) + k_f \Omega_n(s) = k_d I_d(s) - \tau_L(s) \end{cases} \tag{54}$$

where $\Omega_n(s)$, $I_d(s)$, $U(s)$ and $\tau_L(s)$ represent the Laplace transformations of $\omega(t)$, $i_d(t)$, $u(t)$ and $\tau_L(t)$ respectively.



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1 Model in Differential Equation

2 Model in Laplace Equation and Transfer Function

3 Block Diagram - Signal-flow Graph- Mason's Rule

4 Model in State-space Equation

5 Nonlinear Model and Linearization

6 Transient and Steady-State Responses

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3 Block Diagram - Signal-flow Graph- Mason's Rule

- Block Diagram
- Signal Flow Graph
- Mason's Rule



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A system block diagram with one feedback block is depicted as in Fig.1

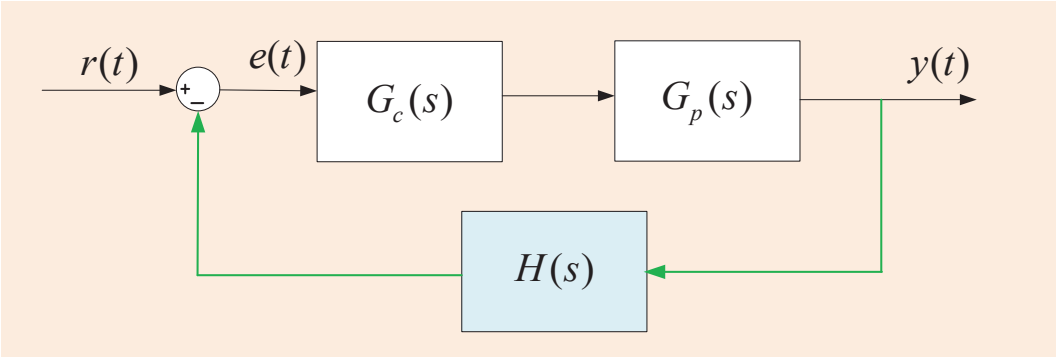


Figure 1: The Typical Closed-loop Block Diagram

The Laplace transformation $E(s)$ of error signal $e(t)$ is

$$E(s) = R(s) - H(s)Y(s) \tag{55}$$

where $R(s)$ and $Y(s)$ are the Laplace transformations of input reference signal $r(t)$ and system output $y(t)$. With the topology connection in Fig.1, one has that

$$Y(s) = G_p(s)G_c(s)E(s) = G_p(s)G_c(s)\left(R(s) - H(s)Y(s)\right) \tag{56}$$

Then, the closed-loop transfer function of system in Fig.1 can be presented as

$$\frac{Y(s)}{R(s)} = \frac{G_c(s)G_p(s)}{1 + G_c(s)G_p(s)H(s)} \tag{57}$$

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In Fig.1, there is only one loop. In engineering, there may exist multiple loops. In Fig.2, a typical system with multiple loops is to be introduced

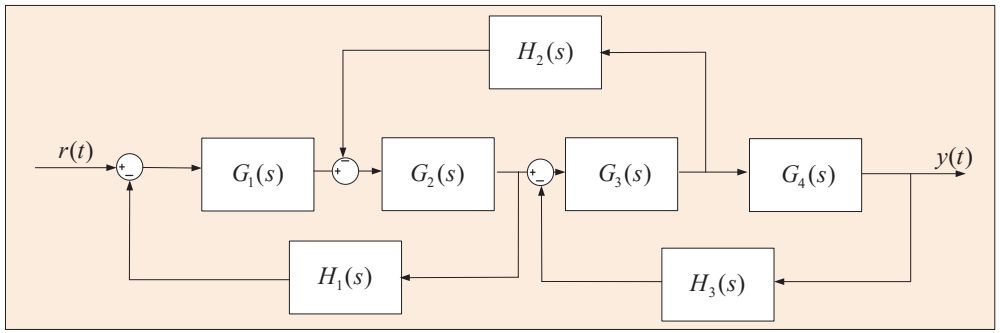


Figure 2: The Typical Closed-loop Block Diagram with Multiple Loops

And, with the equal transformation method, one can transform the system in Fig.2 into Fig.3.

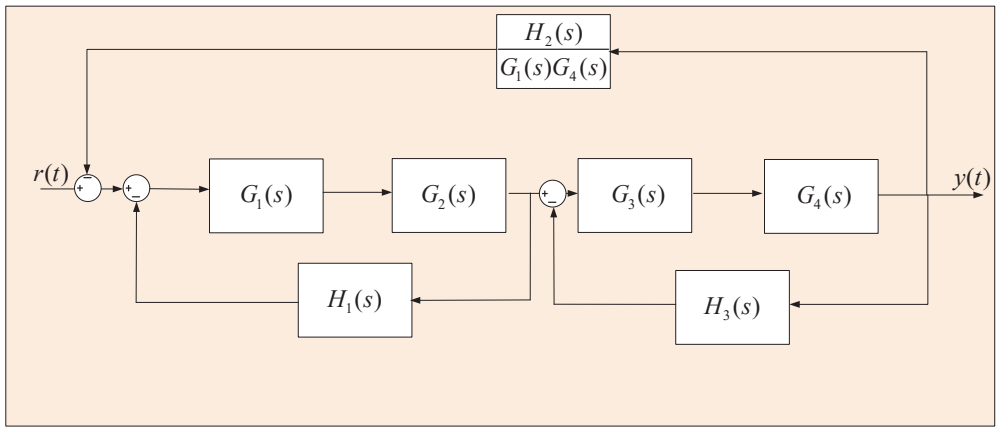


Figure 3: The Typical Closed-loop Block Diagram with Multiple Loops

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Simplify the system in Fig.3 into Fig.4, via equalling the subblocks of the forepart and rear into closed subsystems.

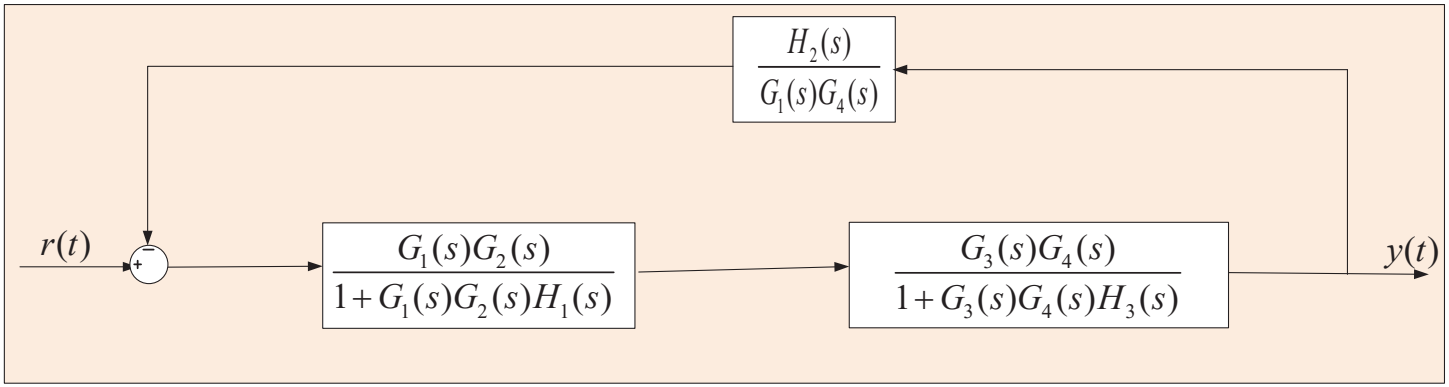


Figure 4: The Typical Closed-loop Block Diagram with Multiple Loops

One can further verify the transfer function below

$$\begin{aligned}\frac{Y(s)}{R(s)} &= \frac{\dfrac{G_1G_2}{1+G_1G_2H_1}\dfrac{G_3G_4}{1+G_3G_4H_3}}{1+\dfrac{H_2}{G_1G_4}\dfrac{G_1G_2}{1+G_1G_2H_1}\dfrac{G_3G_4}{1+G_3G_4H_3}} = \frac{\dfrac{G_1G_2}{1+G_1G_2H_1}\dfrac{G_3G_4}{1+G_3G_4H_3}}{1+H_2\dfrac{G_2}{1+G_1G_2H_1}\dfrac{G_3}{1+G_3G_4H_3}} \\ &= \frac{G_1G_2G_3G_4}{(1+G_1G_2H_1)(1+G_3G_4H_3)+H_2G_2G_3}\end{aligned} \tag{58}$$

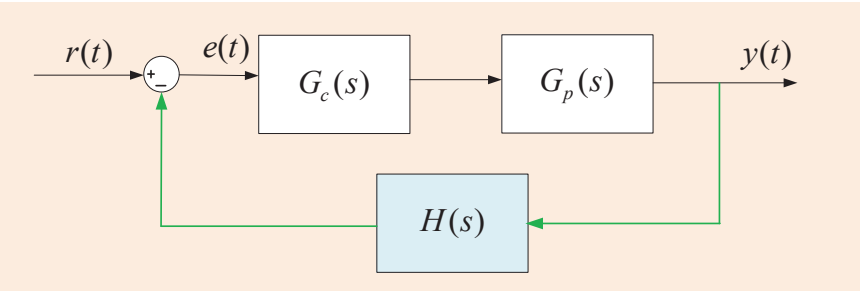
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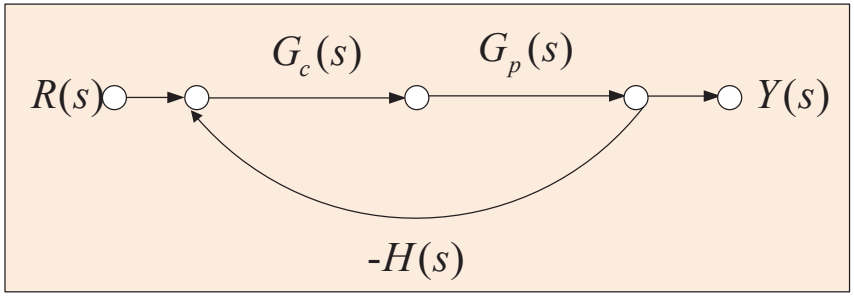
3 Block Diagram - Signal-flow Graph- Mason's Rule

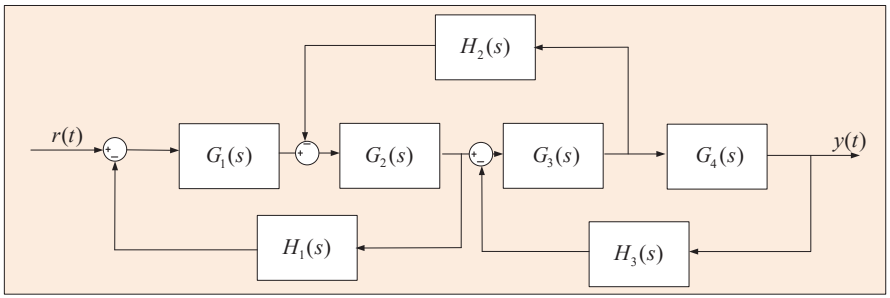
- Block Diagram
- Signal Flow Graph
- Mason's Rule

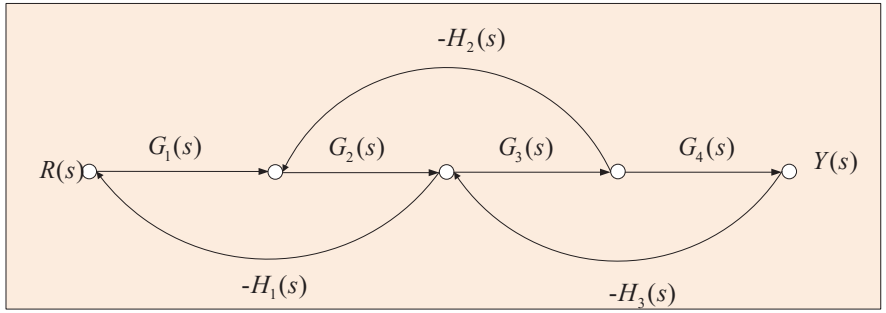
blocks + signals+sum junctions+pickoff points

nodes + direction lines+ signal names









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3 Block Diagram - Signal-flow Graph- Mason's Rule

- Block Diagram
- Signal Flow Graph
- Mason's Rule



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According to the Mason's rule, the transfer function between an input to an output

$$G = \sum_{k=1}^{n} \frac{T_k \Delta_k}{\Delta} \tag{59}$$

G:	from input to output
n:	total number of forward channels from input to output nodes
T_k:	$k-th$ gain of forward path
Δ_k:	cofactor of the $k-th$ forward path

and Δ is

$$\Delta = 1 - \sum L_1 + \sum L_2 - \sum L_3 \cdots \sum (-1)^m L_m \tag{60}$$

L_1:	gain of each feedback loop of the signal flow graph
L_2:	product of any $'2'$ non-contact loop gains
L_3:	product of any $'3'$ non-touching loop gains
L_m:	product of any $'m'$ non-touching loop gain



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Example

Consider the signal flow graph as in Fig.5,

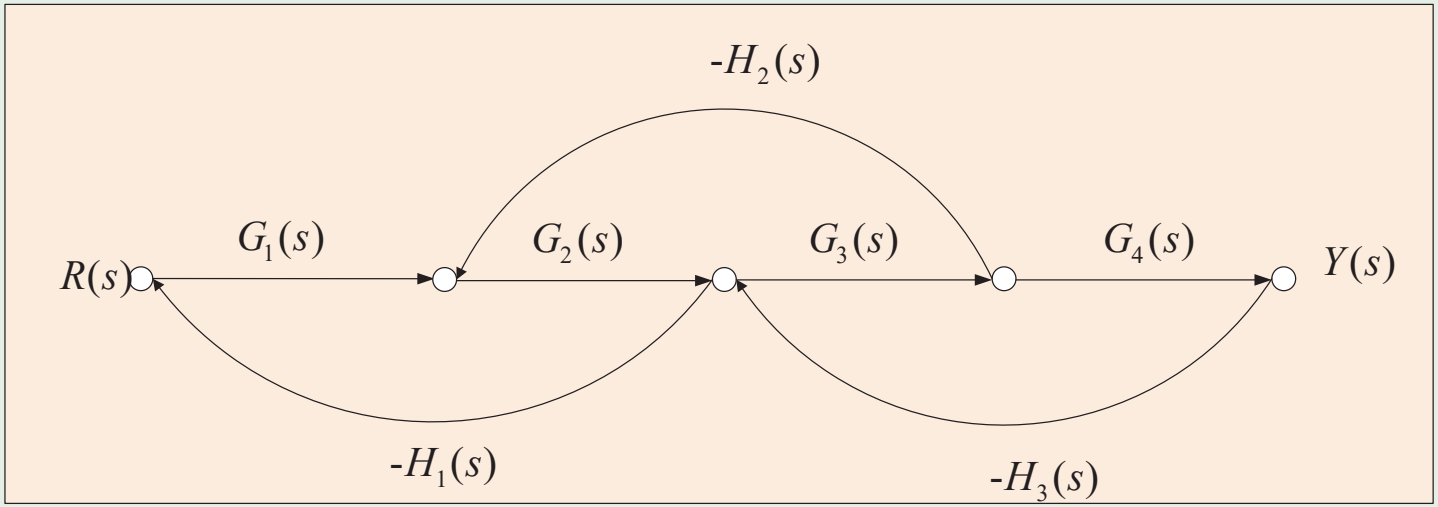


Figure 5: The Typical Closed-loop Signal Flow Graph with Multiple Loops

examine the transfer function via the Mason's rule. One can obtain the system loops as below

$$\begin{array}{lcl} L_{p1} & : & -G_1G_2H_1 \\ L_{p2} & : & -G_2G_3H_2 \\ L_{p3} & : & -G_3G_4H_3 \end{array} \tag{61}$$

Example

one can find that L_{p_1} has no connect with L_{p3} but is connected with L_{p2}, and L_{p2} is connected with L_{p3}, then with the Mason's rule,

$$\begin{array}{rcl} G & = & \sum_{k=1}^{n} \frac{T_k \Delta_k}{\Delta} \\ \Delta & = & 1 - \sum L_1 + \sum L_2 - \sum L_3 \cdots \sum (-1)^m L_m \end{array} \tag{62}$$

one has

$$\begin{array}{rcl} \Delta & = & 1 + [G_1 G_2 H_1 + G_2 G_3 H_2 + G_3 G_4 H_3] \\ & & + (G_1 G_2 H_1)(G_3 G_4 H_3) \end{array} \tag{63}$$

the forward loop is

$$T_1 : G_1 G_2 G_3 G_4 \tag{64}$$

there is no loop not connected with the forward loop, so $\Delta_1 = 1$. The transfer function is

$$G(s) = \frac{T_1 \Delta_1}{\Delta} \tag{65}$$

It equals with the transfer function in (58).



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Example

Consider a complicate signal flow graph in Fig.6, one can find that it's not so convenient and easy to find the transfer function directly, then the Mason's rule helps us much

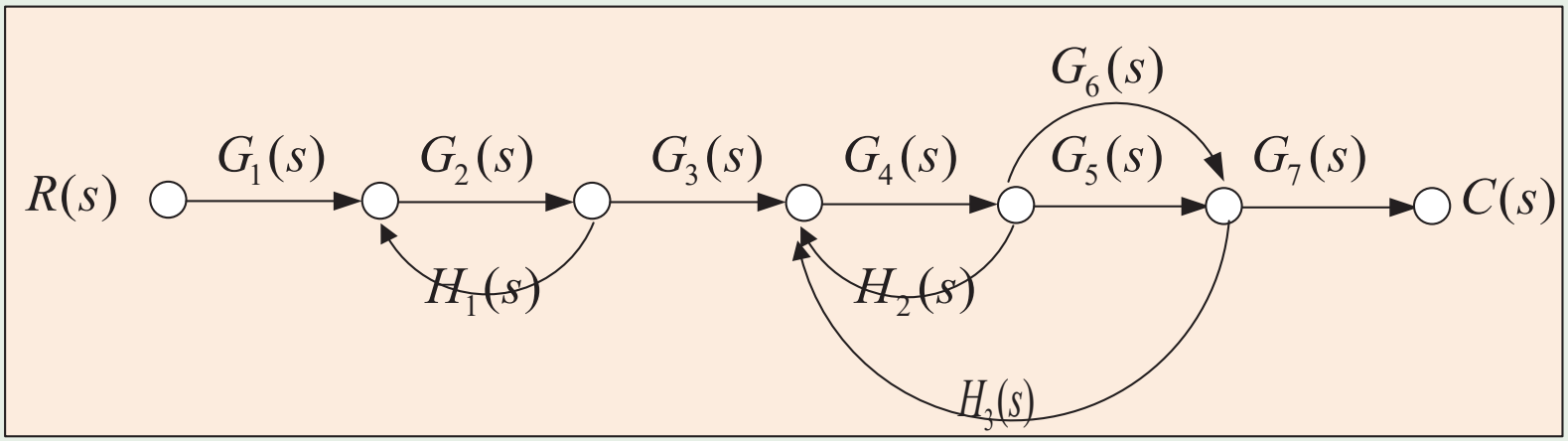


Figure 6: The Typical Signal Flow Graph for Mason's Rule

One can find that there are four loops in this signal graph, which are below

$$\begin{array}{lcl} L_{p1} & : & G_2H_1 \\ L_{p2} & : & G_4H_2 \\ L_{p3} & : & G_4G_5H_3 \\ L_{p4} & : & G_4G_6H_3 \end{array} \tag{66}$$

Example

Forward-path gain is the product of gains found by traversing a path from the input node to the output node in the direction of the signal-flow graph. There are two forward-path gains as follows

$$\begin{array}{lcl} T_1 & : & G_1G_2G_3G_4G_5G_7 \\ T_2 & : & G_1G_2G_3G_4G_6G_7 \end{array} \tag{67}$$

Nontouching loops are loops that do not have any nodes in common. The loop L_{p1} has no connection with L_{p2}, L_{p3} and L_{p4}, it has that

$$\begin{array}{rcl} \Delta & = & 1 - G_2H_1 - G_4H_2 - G_4G_5H_3 - G_4G_6H_3 \\ & & +(G_2H_1)(G_4H_2) + (G_2H_1)(G_4G_5H_3) + (G_2H_1)(G_4G_6H_3) \end{array} \tag{68}$$

$\Delta_1 = 1, \Delta_2 = 1$. According to the Mason's rule, one can obtain the transfer function below

$$G(s) = \frac{T_1 + T_2}{\Delta} \tag{69}$$



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Example

Consider a more complicate signal flow graph in Fig.7,

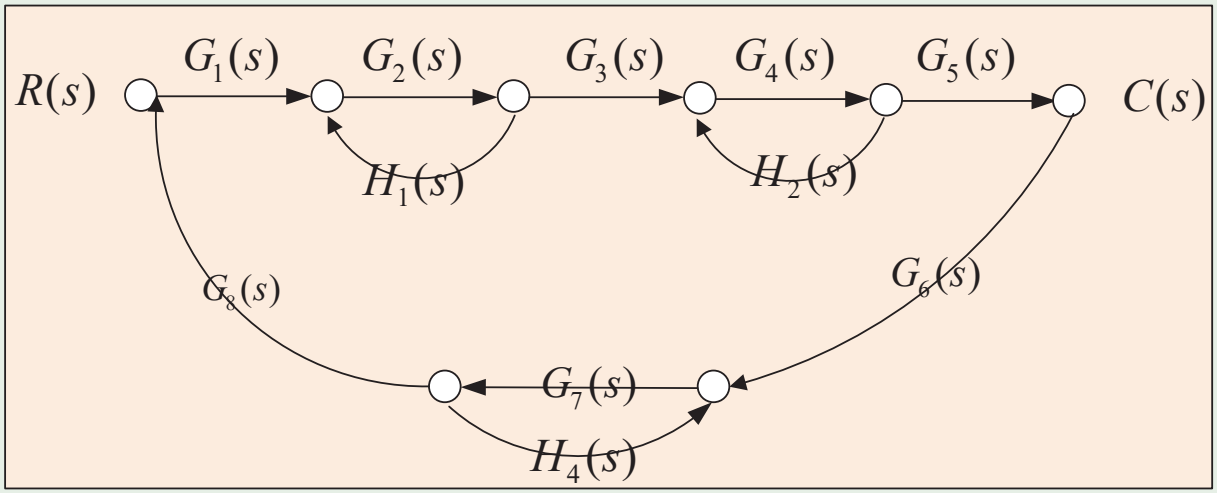


Figure 7: The Typical Signal Flow Graph for Mason's Rule

According to the topology of the signal flow graph, there are four loops, which are

$$\begin{array}{lcl} L_{p1} & : & G_2H_1 \\ L_{p2} & : & G_4H_2 \\ L_{p3} & : & G_7H_4 \\ L_{p4} & : & G_1G_2G_3G_4G_5G_6G_7G_8 \end{array} \tag{70}$$

Example

It verifies that, L_{p1} doesn't touch L_{p2}, L_{p3}, but touches with L_{p4}. L_{p2} doesn't touch L_{p3} but touches with L_{p4}. It obtains that

$$\begin{aligned}\Delta &= 1-[G_2H_1+G_4H_2+G_7H_4+G_2G_3G_4G_5G_6G_7G_8]\\ &\quad +[(G_2H_1)(G_4H_2)+(G_2H_1)(G_7H_4)+(G_4H_2)(G_7H_4)]\\ &\quad -[(G_2H_1)(G_4H_2)(G_7H_4)]\end{aligned} \tag{71}$$

There is only one forward path

$$T_1 : G_1G_2G_3G_4G_5 \tag{72}$$

and one forms Δ_k by eliminating from Δ, those loop gains that touch the $k-th$ forward path below

$$\Delta_1 = 1-G_7H_4 \tag{73}$$

Therefore, one verifies the transfer function below

$$G(s)=\frac{T_1\Delta_1}{\Delta} \tag{74}$$



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Example

Consider the LRC circuits again

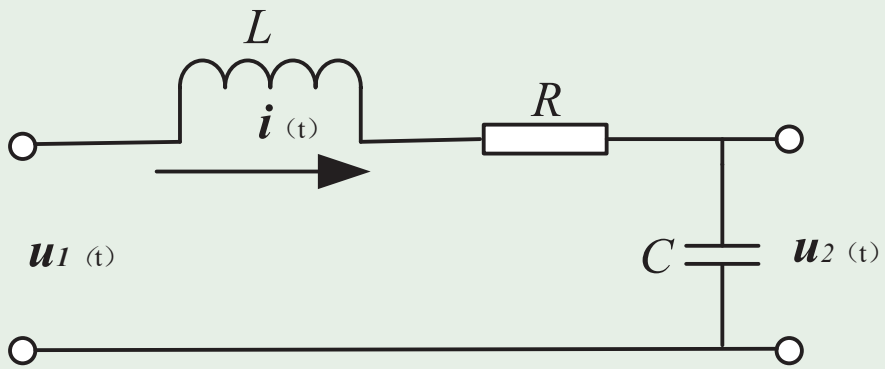


Figure 8: Electrical RLC Circuit

whose differential model is

$$\begin{cases} L\dfrac{di(t)}{dt} &= u_1(t) - u_2(t) - Ri(t) \\ RC\dot{u}_2(t) + LC\ddot{u}_2(t) &= u_1(t) - u_2(t) \end{cases} \tag{75}$$

Example

if one defines that

$$E=\begin{bmatrix} L & 0 & 0 \\ 0 & RC & LC \\ 0 & 1 & 0 \end{bmatrix}, A=\begin{bmatrix} R & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, x(t)=\begin{bmatrix} i(t) \\ u_2(t) \\ \dot{u}_2(t) \end{bmatrix}, B=\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$$

$C=[0\ 1\ 0]$, input $u(t)=u_1(t)$, and the output $y(t)=u_2(t)$, one can verify the equal differential equation as follow

$$\begin{cases} E\dot{x}(t) &= Ax(t)+Bu(t) \\ y(t) &= Cx(t) \end{cases} \tag{76}$$

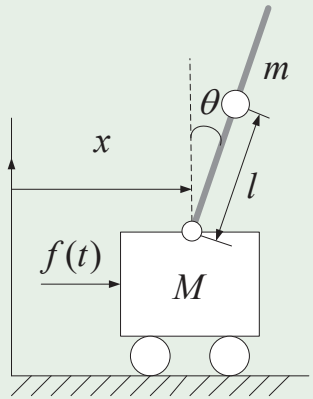
E is nonsingular, whose determinant is nonzero, so the system model can be further verified into norm form as follow

$$\begin{cases} \dot{x}(t) &= \tilde{A}x(t)+\tilde{B}u(t) \\ y(t) &= Cx(t) \end{cases} \tag{77}$$

where $\tilde{A}=E^{-1}A, \tilde{B}=E^{-1}B$.

Example

Consider the cart inverted pendulum



whose differential equations are

$$\begin{cases} M\ddot{x} &= f - m\dfrac{d^2}{dt^2}\left(x + l\sin(\theta)\right) \\ I\ddot{\theta} &= \left(m\dfrac{d^2}{dt^2}\left(l\cos(\theta)\right) + mg\right) l\sin(\theta) - \left(f - M\ddot{x}\right) l\cos(\theta) \end{cases} \tag{78}$$

and

$$\begin{cases} (M+m)\,\ddot{x} + ml\cos(\theta)\ddot{\theta} - ml\sin(\theta)\dot{\theta}^2 &= f \\ ml\cos(\theta)\ddot{x} + \left(I + ml^2\right)\ddot{\theta} - mgl\sin(\theta) &= 0 \end{cases} \tag{79}$$

Example

It can also be presented as

$$\mathcal{M}\ddot{\zeta} + \mathcal{C}(\theta, \dot{\theta})\dot{\zeta} + \mathcal{G}(\theta) = \Gamma f \tag{80}$$

where

$$\begin{aligned}
\mathcal{M} &= \begin{bmatrix} (M+m) & ml\cos(\theta) \\ ml\cos(\theta) & \left(I+ml^2\right) \end{bmatrix}, \zeta = \begin{bmatrix} x \\ \theta \end{bmatrix}, \\
\mathcal{C}(\theta, \dot{\theta}) &= \begin{bmatrix} 0 & -ml\sin(\theta)\dot{\theta} \\ 0 & 0 \end{bmatrix}, \\
\mathcal{G}(\theta) &= \begin{bmatrix} 0 \\ -mgl\sin(\theta) \end{bmatrix}, \quad \Gamma = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.
\end{aligned}$$

Then the system can be expressed as in state space as follow

$$\begin{cases} \dot{\xi}(t) &= \bar{A}(\theta, \dot{\theta})\xi(t) + \bar{G}(\theta) + \bar{B}u(t) \\ y(t) &= \bar{C}\xi(t) \end{cases} \tag{81}$$

where

$$\begin{aligned}
\xi(t) &= \begin{bmatrix} \zeta(t) \\ \dot{\zeta}(t) \end{bmatrix}, \bar{A}(\theta, \dot{\theta}) = \begin{bmatrix} I & 0 \\ 0 & \mathcal{M}^{-1} \end{bmatrix} \begin{bmatrix} 0 & I \\ 0 & -\mathcal{C}(\theta, \dot{\theta}) \end{bmatrix}, \bar{G}(\theta) = \begin{bmatrix} I & 0 \\ 0 & \mathcal{M}^{-1} \end{bmatrix} \begin{bmatrix} 0 \\ -\mathcal{G}(\theta) \end{bmatrix} \\
\bar{B} &= \begin{bmatrix} I & 0 \\ 0 & \mathcal{M}^{-1} \end{bmatrix} \begin{bmatrix} 0 \\ \Gamma \end{bmatrix}, \bar{C} = [I,\ 0], \text{and } u(t) = f(t).
\end{aligned}$$

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5 Nonlinear Model and Linearization
- Approximation Around Equilibrium Points
- Piecewise Linear System



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Consider a nonlinear system below

$$\dot{x}(t) = f(x) \tag{82}$$

where $f(x) \in \mathbb{R}^n \ \forall \ x \in \mathbb{R}^n$.
Function $f(x)$ is locally Lipschitz at point x_0 if it satisfies that

$$\|f(x) - f(y)\| \leq L\|x - y\|, \ \forall \ x, \ y$$

where L is positive and $\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2}$.
At equilibrium point x_0, one has

$$\begin{aligned} \dot{x}(t) &= f(x) \\ &= \left\{ \left[\frac{\partial f}{\partial x} \right]_{x=x_0} \right\} x(t) + \Delta(x)x(t) \end{aligned} \tag{83}$$

the Jacobian matrix $\left[\frac{\partial f}{\partial x}\right] \in \mathbb{R}^{n \times n}$ is

$$\left[\frac{\partial f}{\partial x} \right] = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

where $f(t) = [f_1(t), f_2(t) \cdots f_n(t)]^T$, $x(t) = [x_1(t), x_2(t) \cdots x_n(t)]^T$, then one has

$$\dot{x}(t) = Ax(t) + \Delta(x)x(t) \tag{84}$$

One of the approximation candidate equation can be

$$\dot{x}(t) = Ax(t) \tag{85}$$

the additional condition must hold is that the approximation should be uniform in t, that is

$$\lim_{x \to x_0} \Delta(x) = 0 \tag{86}$$

Example

A simple pendulum in Fig.9, whose input torque $\tau(t)$ is offered by electrical motor. The output is the angle θ. The moment of inertia about the pivot is J, the external force is gravity Mg With the forces applied to the system, and the physical relationships, one has that

$$J\ddot{\theta} = -\tau(t) + MgL\sin(\theta) \tag{87}$$

the model dynamics below

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{-\tau(t)}{J} + \frac{MgL}{J}\sin(x_1) \\ y &= x_1 \end{aligned} \tag{88}$$

Example

The term $\sin(x_1)$ is nonlinear. One can verify the linearized system model below

$$\dot{x} = f(x_0) + \left.\frac{\partial f(x)}{\partial x}\right|_{x_0} (x - x_0) \tag{89}$$

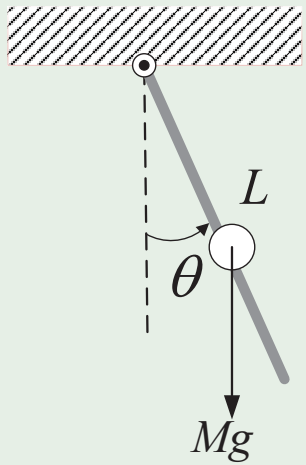


Figure 9: A Simple Pendulum

Example

Then, one has the linearized system model below

Case $x_0 = 0$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{-\tau(t)}{J} + \frac{MgL}{J}x_1 \end{aligned} \tag{90}$$

Case $x_0 = \frac{\pi}{2}$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{-\tau(t)}{J} + 0x_1 \end{aligned} \tag{91}$$

Case $x_0 = \pi$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \frac{-\tau(t)}{J} - \frac{MgL}{J}x_1 \end{aligned} \tag{92}$$



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5 Nonlinear Model and Linearization

- Approximation Around Equilibrium Points
- Piecewise Linear System



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Consider a typical nonlinear system below

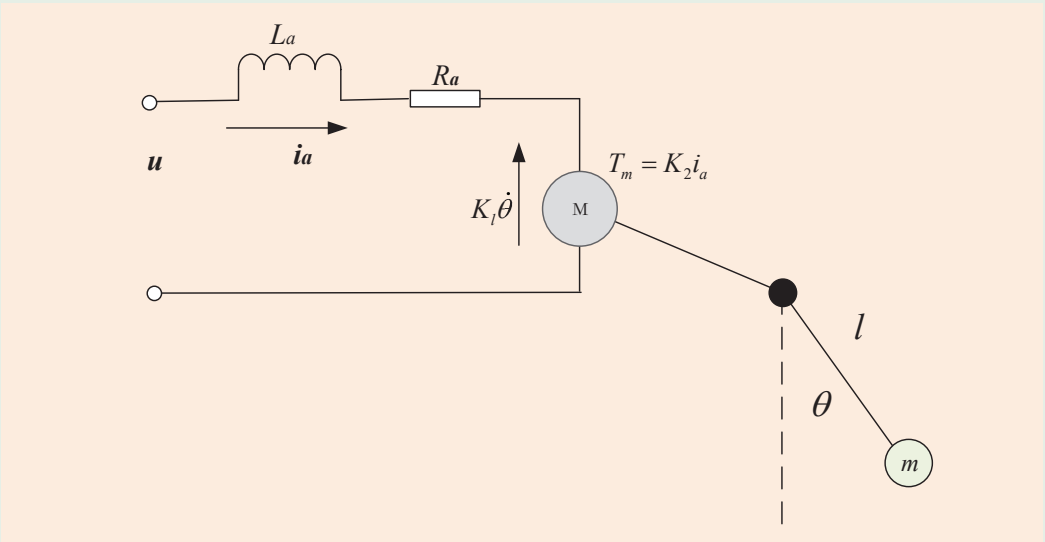
$$\begin{cases} \dot{x} &= f(x,u,t) \\ y &= g(x,u,t) \end{cases} \tag{93}$$

whose piecewise linear model can be presented as

$$\begin{cases} \dot{x} &= A_i x + a_i + B_i u \\ y &= C_i x + c_i + D_i u \end{cases} \quad \text{for } x \in X_i. \tag{94}$$

Example

Consider an electrical-motor driven pendulum system as in following figure,



Example

With Kirchhoff's and Newton's laws, one has the system dynamics

$$\begin{cases} \ddot{\theta}(t) &= -\dfrac{B_\theta}{ml^2+J_m}\dot{\theta}(t) + \dfrac{K_2}{ml^2+J_m}i_a(t) - \dfrac{mgl}{ml^2+J_m}\sin(\theta(t)) \\ \dot{i}_a(t) &= -\dfrac{R_a}{L_a}i_a(t) - \dfrac{K_1}{L_a}\dot{\theta}(t) - \dfrac{1}{L_a}u(t) \end{cases} \tag{95}$$

where $K_1 = 0.052V/rad/sec$, $K_2 = 0.052Nm/A$, $R_a = 1.3\Omega$, $L_a = 1.8 \times 10^{-3}H$, $J_m = 2.02 \times 10^{-5}kgm^2$, $B_\theta = 0.01Nm \cdot \sec$, $l = 0.24m$, and $m = 0.1kg$. Let $x_1(t) = \theta$, $x_2(t) = \dot{\theta}$, $x_3(t)$ equal to the driven current i_a, then one has the nonlinear dynamic model below

$$\begin{cases} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -40.69\sin(x_1(t)) - 1.73x_2(t) + 8.996x_3(t) \\ \dot{x}_3(t) &= -28.89x_2(t) - 722.2x_3(t) + 555.5u(t) \\ y(t) &= x_1(t) \end{cases} \tag{96}$$

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Example

The nonlinear term in the second state equation for $\dot{x}_2(t) = -40.69\sin(x_1(t)) - 1.73x_2(t) + 8.996x_3(t)$ can be modeled as a set of seven piecewise linear function as follow

$$\begin{cases} \dot{x} &= A_i x(t) + B_i \begin{bmatrix} 1 & 0 \\ 0 & u(t) \end{bmatrix} \\ y(t) &= Cx(t) \end{cases} \tag{97}$$

where the values m_i and b_i are given in Table 2 below

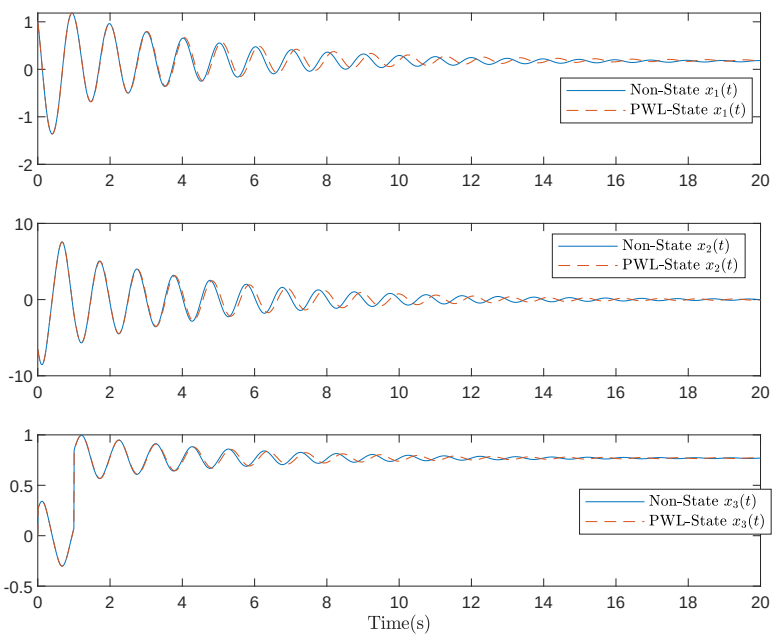
Table 2: Slope and intercept values for $\sin(x_1(t))$

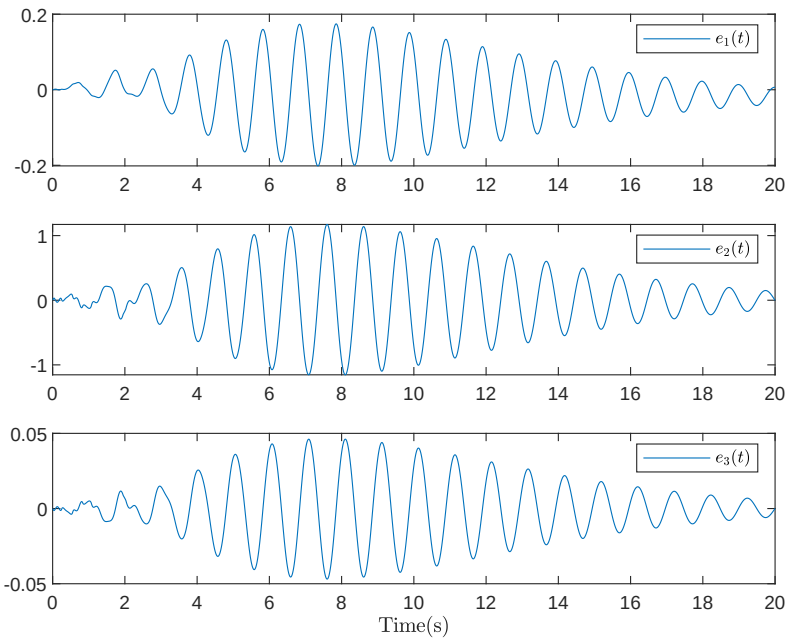
i	m_i	b_i	Range
1	-0.8706	-2.7615	$-\pi \le x_1(t) \le -0.7\pi$
2	-0.3059	-1.5114	$-0.7\pi \le x_1(t) \le -0.5\pi$
3	0.3059	-0.5505	$-0.5\pi \le x_1(t) \le -0.3\pi$
4	0.9107	0	$-0.3\pi \le x_1(t) \le 0.3\pi$
5	0.3059	0.5505	$0.3\pi \le x_1(t) \le 0.5\pi$
6	-0.3059	1.5114	$0.5\pi \le x_1(t) \le 0.7\pi$
7	-0.8706	2.7615	$0.7\pi \le x_1(t) \le \pi$

Example

$$A_i = \begin{bmatrix} 0 & 1 & 0 \\ -40.69m_i & -1.73 & 8.996 \\ 0 & -28.89 & -722.2 \end{bmatrix}, B_i = \begin{bmatrix} 0 & 0 \\ -40.69b_i & 0 \\ 0 & 555.5 \end{bmatrix}, C = [1\ 0\ 0].$$

Initial system states $x(0) = [1\ \ -6.5\ 0]$. Step input reference $u(t)$ begins from 1 second.





During the beginning phase (about 0–5s), the system state error amplitude is low.
In the middle phase, the error amplitude becomes larger.
In the end stage, the curve shape differs and the error amplitude becomes small.

1 Model in Differential Equation

2 Model in Laplace Equation and Transfer Function

3 Block Diagram - Signal-flow Graph- Mason's Rule

4 Model in State-space Equation

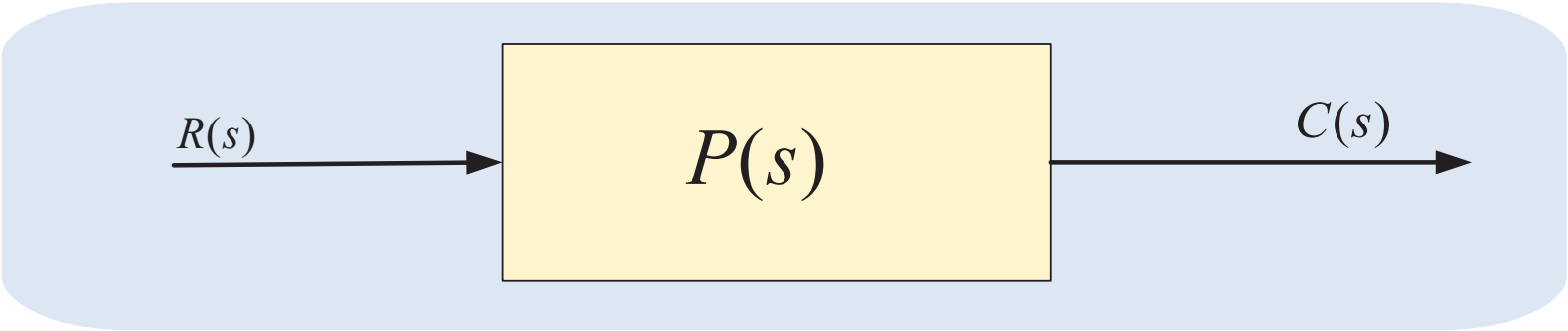
5 Nonlinear Model and Linearization

6 Transient and Steady-State Responses

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6 Transient and Steady-State Responses
- Typical Transient Responses
- Steady-State Error

Consider the typical system $P(s)$ below



The first-order, second order and higher order systems are

$$\begin{aligned}
&\textbf{First-Order System:} && \frac{C(s)}{R(s)} = P(s) = \frac{1}{Ts+1} \\
&\textbf{Second-Order System:} && \frac{C(s)}{R(s)} = P(s) = \frac{\omega_n^2}{s^2+2\eta\omega_n s+\omega_n^2} \\
&\textbf{Higher-Order System:} && \frac{C(s)}{R(s)} = P(s) = \frac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}
\end{aligned} \tag{98}$$

where z_i and p_i are complex, i.e. $z_i \in \mathbb{C}$ and $p_i \in \mathbb{C}$. It's assumed that $c(0) = 0$, $\dot{c}(0) = 0$, $\cdots$,.

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Impulse Responses

The Laplace transformation of unit-impulse signal $r(t) = \delta(t)$ is $R(s) = 1$, then the output response $C(s)$ is

$$C(s) = P(s)R(s) = P(s) \tag{99}$$

for the three kinds of systems, with zero initial states, one can verify that the Laplace response output is the system transfer function itself.

♦ Output response of unit-impulse input for the first-order system:

For the typical first-order system,

$$C(s) = \frac{1}{Ts+1} \tag{100}$$

By the inverse Laplace transformation, one can verify the output response $c(t)$ as

$$c(t) = \mathfrak{L}^{-1}(C(s)) = \mathfrak{L}^{-1}\left[\frac{1}{s+\frac{1}{T}}\right] = e^{-\frac{t}{T}}, \text{ for } t > 0. \tag{101}$$



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♦ Output response of unit-impulse input for the second-order system:

The second-order system responses list in three cases, i.e. underdamped case $\eta \in (0,1)$, critical case $\eta = 1$ and overdamped case $\eta \in (1,\infty)$.

$$C(s) = \frac{\omega_n^2}{s^2 + 2\eta\omega_n s + \omega_n^2} \tag{102}$$

By the inverse Laplace transformation, one can obtain that
Under-damped Case: $\eta \in (0,1)$**:**

$$\begin{aligned} c(t) &= \mathfrak{L}^{-1}(C(s)) \\ &= \mathfrak{L}^{-1}\left[\frac{\omega_n^2}{s^2 + 2\eta\omega_n s + \omega_n^2}\right] \\ &= \frac{\omega_n}{\sqrt{1-\eta^2}} e^{-\eta\omega_n t}\sin(\omega_n\sqrt{1-\eta^2}t) \end{aligned} \tag{103}$$

Critical-damped Case: $\eta = 1$**:**

$$c(t) = \omega_n^2 t e^{-\omega_n t} \tag{104}$$

Over-damped Case: $\eta > 1$**:**

$$c(t) = \frac{\omega_n}{2\sqrt{\eta^2-1}} e^{-(\eta-\sqrt{\eta^2-1})\omega_n t} - \frac{\omega_n}{2\sqrt{\eta^2-1}} e^{-(\eta+\sqrt{\eta^2-1})\omega_n t} \tag{105}$$

Fig.10: unit-impulse responses for typical second-order system under different damped-ratios η and same $\omega_n = 2$. The higher damp ratio η , the faster output response. The lower damp ratio η, the higher overshoot value at about $0.5 - 1s$. And the higher ratio also leads to the decrease of response oscillation.

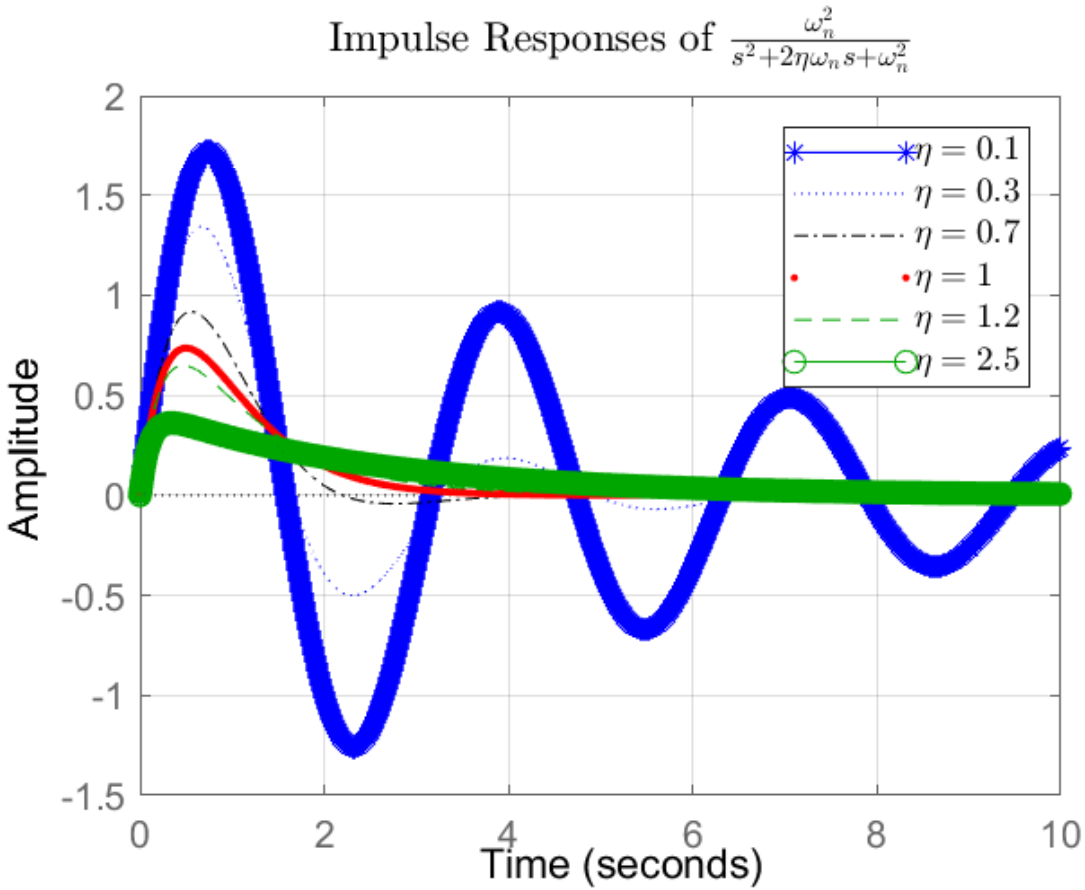


Figure 10: Impulse Responses under Different Damp Ratios

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♦ Output response of unit-impulse input for the higher-order system:

$$C(s) = \frac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)} \tag{106}$$

where $z_i \in \mathbb{C}$, $p_i \in \mathbb{C}$. Decompose the denominator of fractional polynomial function $C(s)$ into the prime form, one has that

$$C(s) = \sum_{i=1}^{q} \frac{\alpha_i}{s+p_i} + \sum_{i=1}^{r} \frac{\beta_i(s+\eta_i\omega_i) + \gamma_i\omega_i\sqrt{1-\eta_i^2}}{s^2+2\eta_i\omega_i s+\omega_i^2} \tag{107}$$

the parameters q and r satisfying that $q+2r=n$. With the Inverse Laplace transformation method, it has that

$$\begin{aligned} c(t) = & \sum_{i=1}^{q} \alpha_i e^{-p_i t} + \sum_{i=1}^{r} \beta_i e^{-\eta_i\omega_i t}\cos(\omega_i\sqrt{1-\eta_i^2}t) \\ & + \sum_{i=1}^{r} \gamma_i e^{-\eta_i\omega_i t}\sin(\omega_i\sqrt{1-\eta_i^2}t) \end{aligned} \tag{108}$$

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Unit-step Responses The Laplace transformation of unit-step response is $R(s) = \frac{1}{s}$, then the output response $C(s)$ is

$$C(s) = P(s)R(s) = \frac{P(s)}{s} \tag{109}$$

♦ Output response of unit-step input for the first-order system:

$$\begin{aligned} C(s) &= \frac{1}{Ts+1}\frac{1}{s} \\ &= \frac{1}{s} - \frac{1}{s+\frac{1}{T}} \end{aligned} \tag{110}$$

By the inverse Laplace transformation, one can verify the output response $c(t)$ as

$$c(t) = \mathfrak{L}^{-1}(C(s)) = \mathfrak{L}^{-1}\left[\frac{1}{s}\right] - \mathfrak{L}^{-1}\left[\frac{1}{s+\frac{1}{T}}\right] = 1 - e^{-\frac{t}{T}}, \text{ for } t > 0. \tag{111}$$



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♦ Output response of unit-step input for the second-order system:

Similarly, the second-order system responses are listed in three cases, i.e. underdamped case, critical case and overdamped case.

1 Underdamped Case: $\eta \in (0,1)$**:**

$$\begin{aligned} C(s) &= \frac{\omega_n^2}{s^2+2\eta\omega_n s+\omega_n^2}\frac{1}{s} \\ &= \frac{\omega_n^2}{(s+\eta\omega_n+j\omega_d)(s+\eta\omega_n-j\omega_d)}\frac{1}{s} \\ &= \frac{1}{s}-\frac{\eta\omega_n}{(s+\eta\omega_n)^2+\omega_d^2}-\frac{s+\eta\omega_n}{(s+\eta\omega_n)^2+\omega_d^2} \end{aligned} \tag{112}$$

where $\omega_d = \omega_n\sqrt{1-\eta^2}$. By the inverse Laplace transformation, one can obtain that

$$\begin{aligned} c(t) &= \mathfrak{L}^{-1}(C(s)) \\ &= \mathfrak{L}^{-1}\left[\frac{1}{s}\right]-\mathfrak{L}^{-1}\left[\frac{\eta\omega_n}{(s+\eta\omega_n)^2+\omega_d^2}\right]-\mathfrak{L}^{-1}\left[\frac{s+\eta\omega_n}{(s+\eta\omega_n)^2+\omega_d^2}\right] \\ &= 1-\frac{\eta}{\sqrt{1-\eta^2}}e^{-\eta\omega_n t}\sin(\omega_d t)-e^{-\eta\omega_n t}\cos(\omega_d t) \end{aligned} \tag{113}$$



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2 Critical-damped Case: $\eta = 1$:

$$\begin{aligned} C(s) &= \frac{\omega_n^2}{s^2 + 2\eta\omega_n s + \omega_n^2}\frac{1}{s} \\ &= \frac{\omega_n^2}{(s+\omega_n)^2}\frac{1}{s} \\ &= \frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2} \end{aligned} \tag{114}$$

By the inverse Laplace transformation, one obtains that

$$c(t) = \mathfrak{L}^{-1}\left[C(s)\right] = 1 - e^{-\omega_n t} - t\omega_n e^{-\omega_n t} \tag{115}$$



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3 Over-damped Case: $\eta > 1$:

$$\begin{aligned} C(s) &= \frac{\omega_n^2}{(s+\eta\omega_n+\omega_n\sqrt{\eta^2-1})(s+\eta\omega_n-\omega_n\sqrt{\eta^2-1})}\frac{1}{s} \\ &= \frac{1}{s}+\left[\frac{1}{2\sqrt{\eta^2-1}(\eta+\sqrt{\eta^2-1})}\right]\left[\frac{1}{s+\eta\omega_n+\sqrt{\eta^2-1}\omega_n}\right] \\ &\quad -\left[\frac{1}{2\sqrt{\eta^2-1}(\eta-\sqrt{\eta^2-1})}\right]\left[\frac{1}{s+\eta\omega_n-\sqrt{\eta^2-1}\omega_n}\right] \end{aligned} \tag{116}$$

By the inverse Laplace transformation, one obtains that

$$\begin{aligned} c(t) &= \mathfrak{L}^{-1}\left[C(s)\right] \\ &= 1+\left[\frac{1}{2\sqrt{\eta^2-1}(\eta+\sqrt{\eta^2-1})}\right]e^{-(\eta\omega_n+\sqrt{\eta^2-1}\omega_n)t} \\ &\quad -\left[\frac{1}{2\sqrt{\eta^2-1}(\eta-\sqrt{\eta^2-1})}\right]e^{-(\eta\omega_n-\sqrt{\eta^2-1}\omega_n)t} \end{aligned} \tag{117}$$



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♦ Output response of step input for the higher-order system:

$$C(s) = \frac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}\frac{1}{s} \tag{118}$$

where $z_i \in \mathbb{C}$, $p_i \in \mathbb{C}$. Decompose the denominator of fractional polynomial function $C(s)$ into the prime form, one has

$$C(s) = \frac{a}{s} + \sum_{i=1}^{q}\frac{\alpha_i}{s+p_i} + \sum_{i=1}^{r}\frac{\beta_i(s+\eta_i\omega_i)+\gamma_i\omega_i\sqrt{1-\eta_i^2}}{s^2+2\eta_i\omega_i s+\omega_i^2} \tag{119}$$

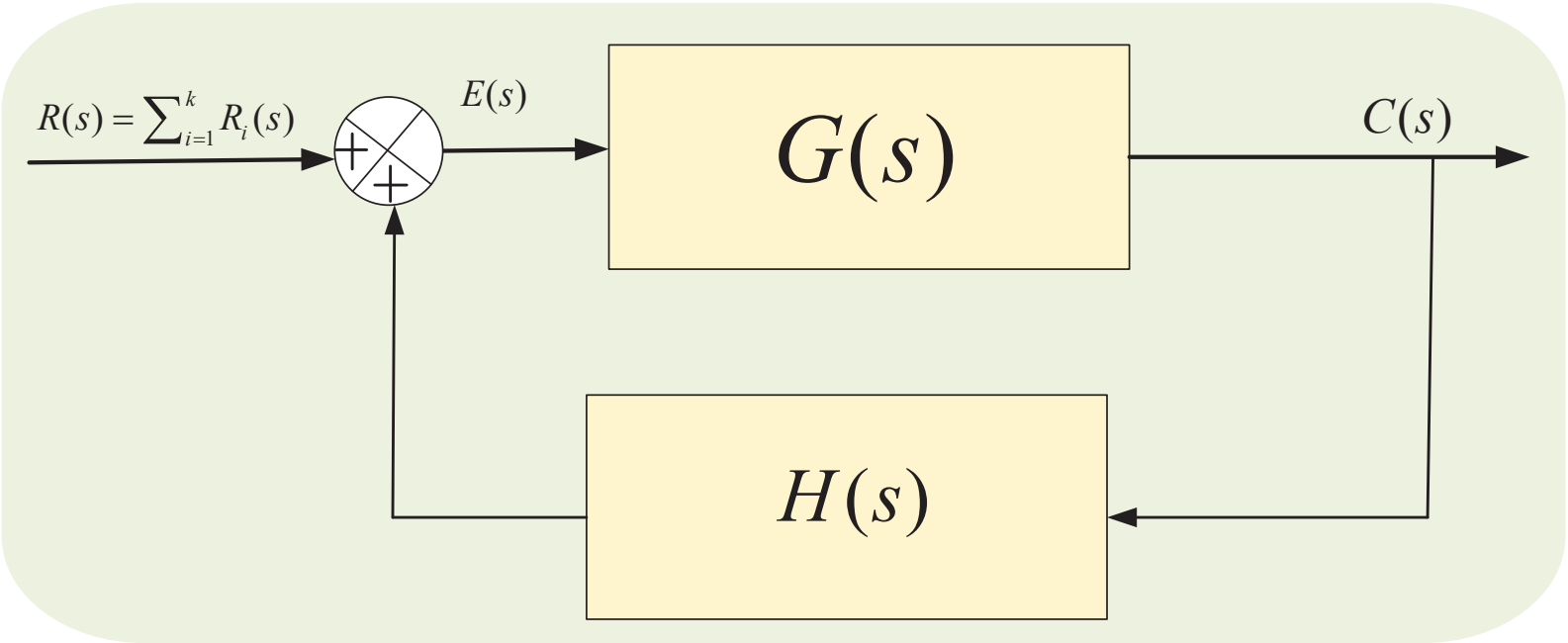
the parameters q and r satisfy that $q+2r=n$. Via the inverse Laplace transformation, one verifies the unit-step responses $c(t)$

$$\begin{aligned} c(t) = {} & a + \sum_{i=1}^{q}\alpha_i e^{-p_i t} + \sum_{i=1}^{r}\beta_i e^{-\eta_i\omega_i t}\cos(\omega_i\sqrt{1-\eta_i^2}\,t) \\ & + \sum_{i=1}^{r}\gamma_i e^{-\eta_i\omega_i t}\sin(\omega_i\sqrt{1-\eta_i^2}\,t) \end{aligned} \tag{120}$$



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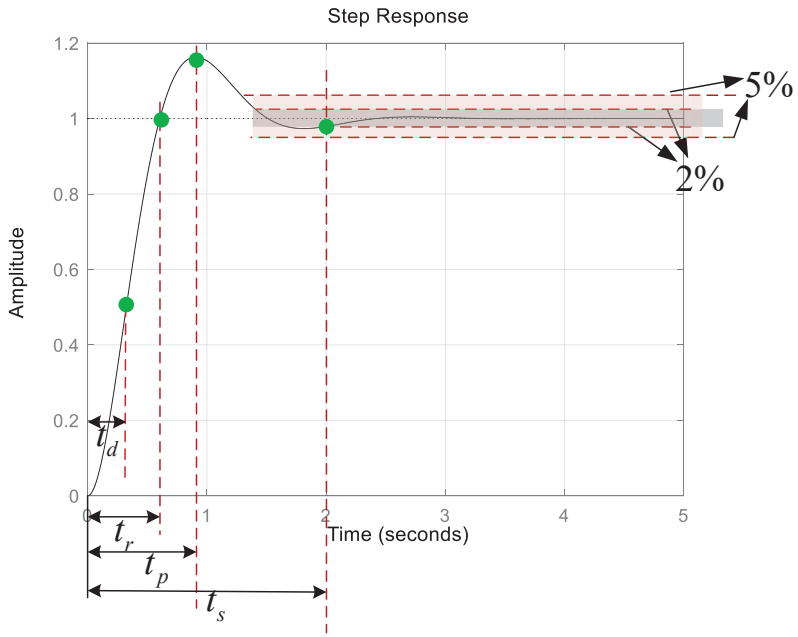
General and Typical Responses



$$c(t) = \mathfrak{L}^{-1}\left[P(s)R(s)\right] = \mathfrak{L}^{-1}\left[P(s)\sum_{i=1}^{k} R_i(s)\right] \tag{121}$$

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Transient-response Specifications



Rise time, t_r:

The time required for the response to rise from one lower to a higher level of its final value. Normally, for under-damped second order systems, from 0% to 100%, while for over-damped systems, from 10% to 90% .

Peak time, t_p:

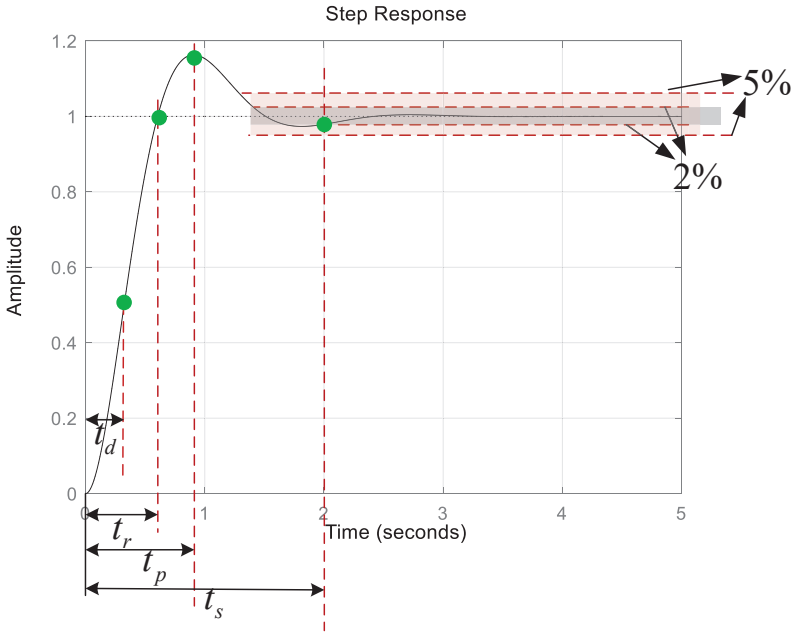
The time required for the response reaching the first peak of the overshoot.

Delay time, t_d:

The delay time indicates the very first time of the response reaching half of the final value.

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Transient-response Specifications



Settling time, t_s:

The time required for the response to reach and stay within a specified range about the final value (usually 2% or 5%), related to the largest time constant of the control system.

Maximum (percent) overshoot, M_p:

The maximum overshoot is the maximum peak value of the response curve measured from unity. If the final steady-state value of the response differs from unity, it is common to use the maximum percent overshoot. Defined as

$$\textit{Maximum percent overshoot} = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\% \tag{122}$$

The amount of the maximum (percent) overshoot directly indicates the relative stability of the system.

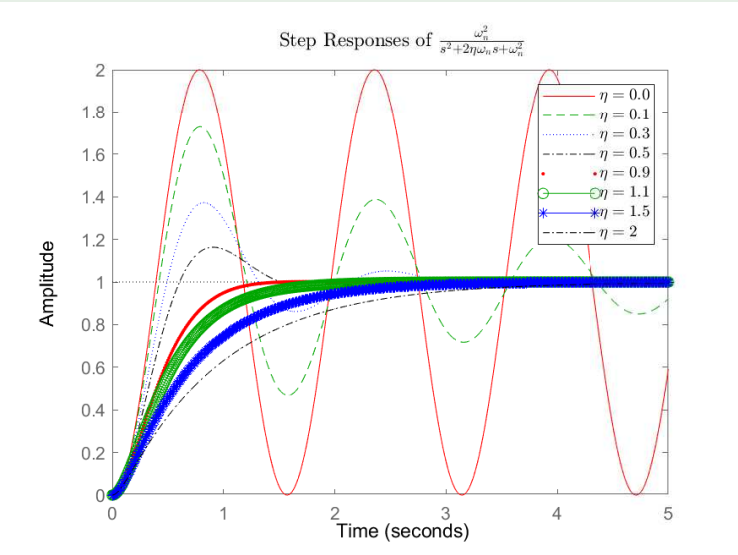
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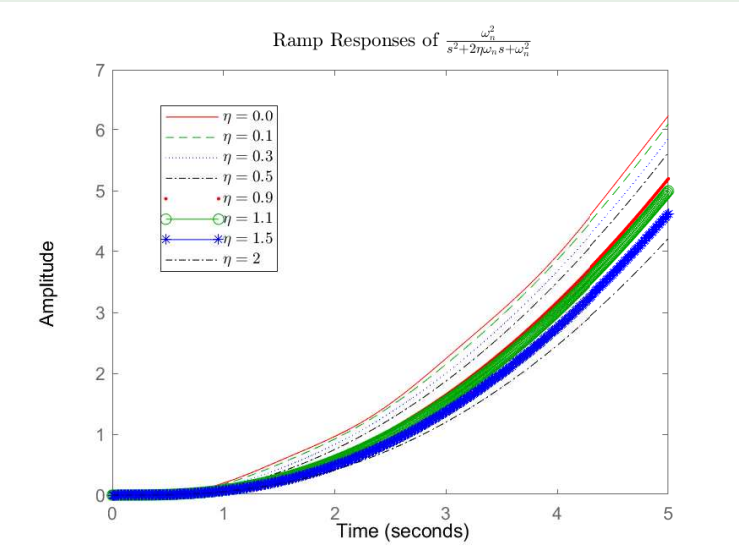
Example

Consider a second order system, whose initial outputs are zero, as follow

$$\frac{\omega_n^2}{s^2 + 2\eta\omega_n s + \omega_n^2} \tag{123}$$

Compare and examine the unit-step and unit-ramp responses under the variations of $\eta = 0, 0.1, 0.3, 0.5, 0.9, 1.1, 1.5$ and 2 with the same $\omega_n = 4$. The state responses can be obtained by the numerical simulation directly.





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6 Transient and Steady-State Responses

- Typical Transient Responses
- Steady-State Error



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Depicted as in Fig.11, the typical negative unit feedback system is given. The transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)} \tag{124}$$

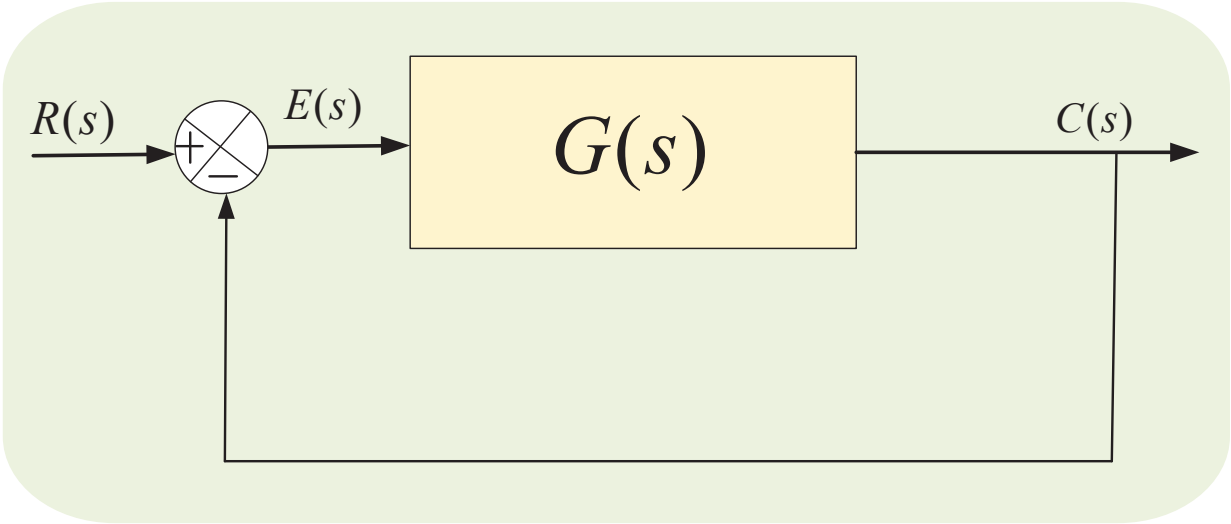


Figure 11: Typical Negative Feedback Control System

Error signal $e(t)$ is the difference between the signal input $r(t)$ and the output $c(t)$, i.e. $e(t) = r(t) - c(t)$. Transfer function from the reference input $r(t)$ to the error signal $e(t)$ is

$$\frac{E(s)}{R(s)} = \frac{R(s) - C(s)}{R(s)} = 1 - \frac{G(s)}{1 + G(s)} = \frac{1}{1 + G(s)} \tag{125}$$

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The error signal $E(s)$ can be expressed as

$$E(s) = \frac{1}{1+G(s)}R(s) \tag{126}$$

According to the **final value theorem**, one obtains the steady state error under reference input

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)} \tag{127}$$

The steady-state error of the system for impulse, unit-step, ramp and acceleration inputs are

$$\begin{aligned}
&\textbf{Impulse input}, && e_{ss} = \lim_{s\to 0} \frac{s}{1+G(s)}, \\
&\textbf{Step input}, && e_{ss} = \lim_{s\to 0} \frac{s}{1+G(s)}\frac{1}{s^1} = \lim_{s\to 0} \frac{1}{1+G(s)}, \\
&\textbf{Ramp input}, && e_{ss} = \lim_{s\to 0} \frac{s}{1+G(s)}\frac{1}{s^2} = \lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s}, \\
&\textbf{Acceleration input}, && e_{ss} = \lim_{s\to 0} \frac{s}{1+G(s)}\frac{1}{s^3} = \lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s^2}.
\end{aligned} \tag{128}$$



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Table 3: Steady-state Errors under Different Input and Limit Cases

Limit Cases	Impulse Input	Step Input	Ramp Input	Acceleration Input
If $\lim_{s\to 0} G(s) = k_1$	$\lim_{s\to 0} \frac{s}{1+G(s)} = 0$	$\lim_{s\to 0} \frac{1}{1+G(s)} = \frac{1}{1+k_1}$	$\lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s} = \infty$	$\lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s^2} = \infty$
If $\lim_{s\to 0} sG(s) = k_2$	$\lim_{s\to 0} \frac{s}{1+G(s)} = 0$	$\lim_{s\to 0} \frac{1}{1+G(s)} = 0$	$\lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s} = \frac{1}{k_2}$	$\lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s^2} = \infty$
If $\lim_{s\to 0} s^2G(s) = k_3$	$\lim_{s\to 0} \frac{s}{1+G(s)} = 0$	$\lim_{s\to 0} \frac{1}{1+G(s)} = 0$	$\lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s} = 0$	$\lim_{s\to 0} \frac{1}{1+G(s)}\frac{1}{s^2} = \frac{1}{k_3}$

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